KCE PATHWAYS · GRADE 3 · UNIT 5: DATA AND DESIGN · WEEK 21 · MATH

Saturday at Pike Place Market — Teacher Lesson Plan & Answer Key

Learning objectives: Students read picture graphs by skip-counting with the key (including half pictures), read scaled bar graphs that count by 2, 5 or 10 (including bars that stop halfway between grid lines), solve how-many-more and how-many-fewer problems, choose a key or scale to make their own graph, and solve two-step graph problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Saturday at Pike Place Market in Seattle, Washington, with Ms. Akana's flower stall, Mr. Bashir's fruit stand, Arlo, Beatriz, Bodhi and Anouk.
2. Warm-Up: Elapsed Time Review (10 pts, Unit 4 skill): finding an end time, a start time or how long an event lasts. If a student misses both, model "jump to the next hour first, then jump the rest of the way."
3. Picture It: Saturday at Pike Place Market. The helpers collect data about what the stalls sell and turn it into graphs (the Market opened August 17, 1907; Rachel the bronze pig has stood there since 1986). Ask: what does a good graph let a seller see in one quick look?
4. Word Wrangler: 6 vocab cards (data, picture graph, key, bar graph, scale, compare).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W21.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Reading Picture Graphs with a Key (10 pts): reading a row and finding a total. Tip: read the key first, then skip-count by it; a half picture is half of the key.
7. Reading Bar Graphs with a Scale (10 pts): reading bars and adding two bars. Tip: read the scale first, run a finger across from the top of the bar, and a bar halfway between two lines is the number in the middle.
8. How Many More? How Many Fewer? (10 pts): subtract the smaller amount from the bigger one, or count the extra pictures and multiply by the key.
9. Make Your Own Graph (10 pts): choosing how many pictures to draw for a key, where a bar stops for a scale, and the best key. Tip: divide the amount by the key (40 ÷ 5 = 8 pictures).
10. Market Challenge: Two-Step Problems (10 pts): read the graph values first, then add or subtract. Ask the student to name step 1 before solving.
11. Bonus Practice: The Lunch Rush (XP only, not scored).
12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches graph vocabulary to meanings, Set 2 matches pictures with a key or a bar position to the number shown, Set 3 matches comparisons to how many more or fewer.

13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people, places, days and months, end mark, commas between a city and a state and in a list, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: The Market Report (10 pts, 5 each): reading and comparing bars or picture rows, a find-the-mistake item and a which-two-together item.
16. Boss Battle, Part 2: Closing Time (10 pts, 5 each): two-step goal problems and a find-the-mistake item about the scale (for example, the 5th grid line is 25 when the lines count by 5).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W21.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Market Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Market Graph Hero!” with the congratulations message “Market graph hero! You turned a whole Saturday of data into graphs!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every graph maker gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: ELAPSED TIME REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: The Market's flower stalls open at 8:45 a.m. Ms. Akana finishes setting up her flower buckets 40 minutes later. What time is it then? Type it like 9:10 a.m.	**9:25 a.m.** (also accepted: 9:25, 9:25 am, 9:25am, 9:25a.m., 9:25 a.m, 9:25 a. m.)	Jump from 8:45 to 9:00: 15 minutes. 40 − 15 = 25 minutes left. 9:00 + 25 minutes = 9:25 a.m.
w2_s0	Set 1: Which explanation is correct? Arlo's bus leaves at 10:50 a.m. and gets to the Market at 11:20 a.m. How long is the ride? [10:50 to 11:00 is 10 minutes, and 11:00 to 11:20 is 20 minutes. 10 + 20 = 30 minutes. / 1120 − 1050 = 70, so the ride is 70 minutes. / The ride is 1 hour 30 minutes, because the hour changed. / The ride is 20 minutes, because the minute hand ends on the 4.]	**10:50 to 11:00 is 10 minutes, and 11:00 to 11:20 is 20 minutes. 10 + 20 = 30 minutes.**	Jump to the next hour first: 10:50 to 11:00 is 10 minutes. Then 11:00 to 11:20 is 20 minutes. 10 + 20 = 30 minutes. An hour has 60 minutes, not 100.
w1_s1	Set 2: The fish sellers toss fish for a crowd from 1:35 p.m. to 2:10 p.m. How many minutes is that?	**35 minutes** (also accepted: 35, 35 minute, 35 min, 35 mins, 35min, 35mins)	1:35 to 2:00 is 25 minutes. 2:00 to 2:10 is 10 minutes. 25 + 10 = 35 minutes.
w2_s1	Set 2: Beatriz's cooking class at the Market ends at 4:15 p.m. The class lasts 55 minutes. What time did it start? Type it like 3:05 p.m.	**3:20 p.m.** (also accepted: 3:20, 3:20 pm, 3:20pm, 3:20p.m., 3:20 p.m, 3:20 p. m.)	To find a start time, jump backward. 4:15 back 15 minutes is 4:00. 55 − 15 = 40 more minutes. 4:00 back 40 minutes is 3:20 p.m.

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: Bodhi helps at the fruit stand from 7:35 a.m. to 8:50 a.m. How long does he help? Type the minutes, or hours and minutes.	**75 minutes** (also accepted: 75, 75 minute, 75 min, 75 mins, 75min, 75mins, 1 hour 15 minutes, 1 hour and 15 minutes, 1 hour, 15 minutes, 1 hr 15 min, 1 h 15 min, 1 hr 15 mins)	7:35 to 8:00 is 25 minutes. 8:00 to 8:50 is 50 minutes. 25 + 50 = 75 minutes, which is 1 hour 15 minutes.
w2_s2	Set 3: Find the mistake. Anouk says 11:45 a.m. to 12:30 p.m. is 1 hour 15 minutes. What went wrong? [She counted 11 to 12 as a whole hour. 11:45 to 12:00 is only 15 minutes, and 12:00 to 12:30 is 30 minutes, so it is 45 minutes. / Nothing is wrong. Going from 11 to 12 always takes 1 hour. / It is really 1 hour 45 minutes, because 45 + 60 = 105. / You cannot find elapsed time when the time goes past noon.]	**She counted 11 to 12 as a whole hour. 11:45 to 12:00 is only 15 minutes, and 12:00 to 12:30 is 30 minutes, so it is 45 minutes.**	11:45 to 12:00 is 15 minutes. 12:00 to 12:30 is 30 minutes. 15 + 30 = 45 minutes, less than 1 hour.

READING PICTURE GRAPHS WITH A KEY (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p1_s0	Set 1: Use the salmon picture graph. How many coho salmon were sold?	**45 salmon** (also accepted: 45 coho salmon, 45 fish, 45)	Each fish picture stands for 10. The coho row has 4 whole fish = 40 and a half fish = 5. 40 + 5 = 45 salmon.
p2_s0	Set 1: Find the mistake. Arlo says the fish market sold 5 sockeye salmon, because the sockeye row has 5 fish pictures. What went wrong? [He forgot the key. Each picture stands for 10 salmon, so 5 pictures show 50 sockeye salmon. / Nothing is wrong. 5 pictures means 5 salmon. / Each picture stands for 5, so it is 25 salmon. / He should add the key: 5 + 10 = 15 salmon.]	**He forgot the key. Each picture stands for 10 salmon, so 5 pictures show 50 sockeye salmon.**	Always read the key. 5 fish pictures, each worth 10: 10, 20, 30, 40, 50. That is 50 sockeye salmon.
p1_s1	Set 2: Use the bouquet picture graph. How many peony bouquets were sold?	**7 bouquets** (also accepted: 7 peony bouquets, 7, seven, seven bouquets, seven peony bouquets)	Each flower stands for 2. The peony row has 3 whole flowers = 6 and a half flower = 1. 6 + 1 = 7 bouquets.
p2_s1	Set 2: Use the same graph. How many bouquets were sold in all, counting all four rows?	**34 bouquets** (also accepted: 34)	Tulips 6 × 2 = 12. Peonies 7. Dahlias 5 × 2 = 10. Lilies 2 and a half flowers = 5. 12 + 7 + 10 + 5 = 34 bouquets.
p1_s2	Set 3: Use the apple picture graph. How many boxes of Fuji apples were sold?	**15 boxes** (also accepted: 15 boxes of Fuji apples, 15)	Each apple stands for 5 boxes. The Fuji row has 3 apples: 5, 10, 15. That is 15 boxes.
p2_s2	Set 3: Which explanation is correct? Why does the Gala row show 30 boxes? [There are 6 apple pictures, and each one stands for 5 boxes: 6 × 5 = 30. / There are 6 apple pictures, so it shows 6 boxes. / There are 6 pictures and the key is 5, so 6 + 5 = 11 boxes. / Each apple stands for 10 boxes, so 6 apples show 60 boxes.]	**There are 6 apple pictures, and each one stands for 5 boxes: 6 × 5 = 30.**	Skip-count by the key for each picture: 5, 10, 15, 20, 25, 30. So 6 × 5 = 30 boxes.

READING BAR GRAPHS WITH A SCALE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: Use the information booth bar graph. How many visitors came at 10 a.m.?	**35 visitors** (also accepted: 35 people, 35)	The scale counts by 10. The 10 a.m. bar stops halfway between 30 and 40, and halfway is 35 visitors.
b2_s0	Set 1: Use the same graph. Which two hours together had exactly 80 visitors? [10 a.m. and noon / 9 a.m. and 11 a.m. / 11 a.m. and noon / 9 a.m. and noon]	**10 a.m. and noon**	10 a.m. had 35 and noon had 45. 35 + 45 = 80. The other pairs make 70, 95, and 65.
b1_s1	Set 2: Use the honey bar graph. How many jars of blackberry honey were sold?	**11 jars** (also accepted: 11 jars of honey, 11)	The scale counts by 2. The blackberry bar stops halfway between 10 and 12, and halfway is 11 jars.
b2_s1	Set 2: Find the mistake. Bodhi says 7 jars of wildflower honey were sold, because the bar goes just past the 6th grid line. What went wrong? [Each grid line is worth 2, not 1. The bar stops halfway between 12 and 14, so it shows 13 jars. / Nothing is wrong. The bar is past the 6th line, so it is 7 jars. / The bar shows 14 jars, because you always round up. / The bar shows 6 jars, because it is near the 6th line.]	**Each grid line is worth 2, not 1. The bar stops halfway between 12 and 14, so it shows 13 jars.**	Read the scale: the lines count 2, 4, 6, 8, 10, 12, 14. The wildflower bar stops halfway between 12 and 14, which is 13.
b1_s2	Set 3: Use the berry bar graph. How many pints of strawberries were sold?	**45 pints** (also accepted: 45 pints of strawberries, 45)	The scale counts by 5. The strawberry bar stops on the 45 line, so 45 pints were sold.
b2_s2	Set 3: Use the same graph. How many pints of raspberries and blackberries were sold together?	**50 pints** (also accepted: 50)	Raspberries: 35 pints. Blackberries: 15 pints. 35 + 15 = 50 pints.

HOW MANY MORE? HOW MANY FEWER? (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Use the postcard picture graph. How many more Space Needle postcards than Mount Rainier postcards were sold?	**25 postcards** (also accepted: 25 more, 25 more postcards, 25)	Space Needle: 9 × 5 = 45. Mount Rainier: 4 × 5 = 20. 45 − 20 = 25 more. Or: 5 more pictures × 5 = 25.
c2_s0	Set 1: Which explanation is correct? How many fewer Fish Toss postcards than Rachel the Pig postcards were sold? [Rachel the Pig has 1 more picture than Fish Toss, and each picture stands for 5, so 5 fewer. / 1 fewer, because Fish Toss has 1 less picture. / 65 fewer, because 35 + 30 = 65. / 6 fewer, because 7 − 6 = 1 and 1 + 5 = 6.]	**Rachel the Pig has 1 more picture than Fish Toss, and each picture stands for 5, so 5 fewer.**	Rachel the Pig: 7 × 5 = 35. Fish Toss: 6 × 5 = 30. 35 − 30 = 5 fewer.
c1_s1	Set 2: Use the customer bar graph. How many more customers visited the fish stall than the bakery?	**45 customers** (also accepted: 45 more, 45 more customers, 45)	Fish: halfway between 80 and 90 is 85. Bakery: 40. 85 − 40 = 45 more customers.
c2_s1	Set 2: Use the same graph. How many fewer customers visited the flower stall than the fruit stand?	**15 customers** (also accepted: 15 fewer, 15 fewer customers, 15)	Fruit: halfway between 70 and 80 is 75. Flowers: 60. 75 − 60 = 15 fewer customers.

ID	Question	Answer	Why (explain to student)
c1_s2	Set 3: Use the bakery picture graph. How many more sourdough loaves than seeded loaves were sold?	**10 loaves** (also accepted: 10 more, 10 more loaves, 10, ten, ten loaves, ten more, ten more loaves)	Sourdough: 5 and a half pictures = 55. Seeded: 4 and a half pictures = 45. 55 − 45 = 10 more loaves.
c2_s2	Set 3: Find the mistake. Anouk says the bakery sold 3 and a half more sourdough loaves than braided loaves. What went wrong? [She counted pictures, not loaves. 3 whole pictures is 30 and half a picture is 5, so 35 more loaves. / Nothing is wrong. There are 3 and a half more pictures, so 3 and a half more loaves. / It is 75 more, because 55 + 20 = 75. / It is 30 more, because the half picture does not count.]	**She counted pictures, not loaves. 3 whole pictures is 30 and half a picture is 5, so 35 more loaves.**	Sourdough 55 − braided 20 = 35. Or use the pictures: 3 and a half more pictures, each worth 10, is 35.

MAKE YOUR OWN GRAPH (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m1_s0	Set 1: Ms. Akana sold 40 bunches of sunflowers. On her picture graph, each flower picture stands for 5 bunches. How many flower pictures should she draw?	**8 pictures** (also accepted: 8 flowers, 8 flower pictures, 8, eight, eight pictures, eight flowers, eight flower pictures)	40 ÷ 5 = 8, because 8 × 5 = 40. She draws 8 pictures.
m2_s0	Set 1: Arlo sold 35 bags of cherries. His key says each picture stands for 10 bags. What should he draw? [3 whole pictures and half of a picture / 35 pictures / 3 pictures / 4 pictures]	**3 whole pictures and half of a picture**	3 pictures make 30. 5 more bags is half of 10, so he adds half a picture: 30 + 5 = 35.
m1_s1	Set 2: Beatriz is making a bar graph with a scale of 2. The honey stand sold 17 jars on Friday. Her bar stops halfway between which two grid lines? Type the two numbers, like 20 and 22.	**16 and 18** (also accepted: 16 and 18 jars, between 16 and 18, 16, 18, 16,18, 16-18, 16 to 18, 18 and 16, sixteen and eighteen)	With a scale of 2 the lines are 14, 16, 18, 20. 17 is between 16 and 18, right in the middle.
m2_s1	Set 2: Bodhi counted 60 tulips, 90 roses, and 40 lilies. Which key makes the shortest picture graph that still uses only whole pictures? [Each picture = 10 flowers / Each picture = 2 flowers / Each picture = 5 flowers / Each picture = 20 flowers]	**Each picture = 10 flowers**	With a key of 10: 6, 9, and 4 whole pictures, 19 in all. A key of 2 or 5 needs many more pictures. A key of 20 would need 4 and a half pictures for 90 roses.
m1_s2	Set 3: Mr. Bashir wants to show 14 crates of peaches. His key says each picture stands for 2 crates. How many pictures does he draw?	**7 pictures** (also accepted: 7, seven, seven pictures)	14 ÷ 2 = 7, because 7 × 2 = 14. He draws 7 pictures.
m2_s2	Set 3: Find the mistake. Bodhi's key says each picture stands for 5 pounds. To show 20 pounds of plums, he drew 20 pictures. What went wrong? [Each picture already stands for 5 pounds, so he needs only 20 ÷ 5 = 4 pictures. / Nothing is wrong. 20 pounds needs 20 pictures. / He needs 25 pictures, because 20 + 5 = 25. / He needs 100 pictures, because 20 × 5 = 100.]	**Each picture already stands for 5 pounds, so he needs only 20 ÷ 5 = 4 pictures.**	4 pictures × 5 pounds = 20 pounds. Drawing 20 pictures would show 100 pounds.

MARKET CHALLENGE: TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Two steps! Ms. Akana's picture graph uses a key where each flower stands for 5 bouquets. Tulips show 7 flowers and daisies show 4 flowers. At closing time, 12 of those bouquets are still unsold. How many tulip and daisy bouquets did she sell?	**43 bouquets** (also accepted: 43)	Step 1: tulips 7 × 5 = 35 and daisies 4 × 5 = 20, so 35 + 20 = 55 bouquets. Step 2: 55 − 12 = 43 bouquets sold.
t2_s0	Set 1: On Beatriz's honey bar graph (scale of 2), the blackberry bar stops halfway between 14 and 16, and the clover bar stops on the 8 line. She wants each kind to reach 20 jars. How many more jars must she sell in all?	**17 jars** (also accepted: 17 more jars, 17)	Blackberry is 15 jars, so it needs 20 − 15 = 5 more. Clover is 8 jars, so it needs 20 − 8 = 12 more. 5 + 12 = 17 more jars.
t1_s1	Set 2: Two steps! The fish market's bar graph (scale of 10) shows 45 salmon sold on Friday and 70 on Saturday. On Sunday the market sold 20 fewer salmon than Friday and Saturday together. How many salmon were sold on Sunday?	**95 salmon** (also accepted: 95 fish, 95)	Step 1: 45 + 70 = 115 salmon on Friday and Saturday. Step 2: 115 − 20 = 95 salmon on Sunday.
t2_s1	Set 2: Mr. Bashir's picture graph uses a key where each picture stands for 2 crates. The peach row has 9 pictures and the plum row has 6 pictures. He sold 5 crates of peaches and 5 crates of plums. How many crates are left in all?	**20 crates** (also accepted: 20)	Step 1: peaches 9 × 2 = 18 and plums 6 × 2 = 12, so 18 + 12 = 30 crates. Step 2: he sold 5 + 5 = 10, and 30 − 10 = 20 crates left.
t1_s2	Set 3: Anouk's picture graph uses a key where each fish stands for 10 pounds. The halibut row shows 3 and a half fish, and the cod row shows 6 fish. How many fewer pounds of halibut than cod were sold?	**25 pounds** (also accepted: 25 fewer, 25 fewer pounds, 25)	Step 1: halibut 3 and a half fish = 35 pounds, and cod 6 fish = 60 pounds. Step 2: 60 − 35 = 25 fewer pounds.
t2_s2	Set 3: Which explanation is correct? Arlo counted 25 kids at Rachel the Pig before lunch and 15 kids after lunch. He wants one picture graph row for all of them, with each picture standing for 5 kids. How many pictures does he draw? [Add first: 25 + 15 = 40 kids. Then 40 ÷ 5 = 8 pictures. / 25 ÷ 5 = 5 pictures, so 5 pictures is all he needs. / 25 + 15 = 40, so he draws 40 pictures. / 25 − 15 = 10, so he draws 10 pictures.]	**Add first: 25 + 15 = 40 kids. Then 40 ÷ 5 = 8 pictures.**	Step 1: find the total, 25 + 15 = 40 kids. Step 2: each picture is 5 kids, and 8 × 5 = 40, so he draws 8 pictures.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match key	**tells how many each picture stands for**	2 points if matched on the first try.
match_s0_2	Set 1: Match scale	**the number the grid lines count by**	2 points if matched on the first try.
match_s0_3	Set 1: Match data	**facts or numbers we collect**	2 points if matched on the first try.
match_s0_4	Set 1: Match picture graph	**uses pictures to show data**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s0_5	Set 1: Match bar graph	**uses bars to show data**	2 points if matched on the first try.
match_s1_1	Set 2: Match 4 pictures, key of 5	**20**	2 points if matched on the first try.
match_s1_2	Set 2: Match 3½ pictures, key of 10	**35**	2 points if matched on the first try.
match_s1_3	Set 2: Match 6 pictures, key of 2	**12**	2 points if matched on the first try.
match_s1_4	Set 2: Match bar halfway between 40 and 50	**45**	2 points if matched on the first try.
match_s1_5	Set 2: Match bar halfway between 6 and 8	**7**	2 points if matched on the first try.
match_s2_1	Set 3: Match 60 compared with 25	**35 more**	2 points if matched on the first try.
match_s2_2	Set 3: Match 2 more pictures, key of 10	**20 more**	2 points if matched on the first try.
match_s2_3	Set 3: Match 3 fewer pictures, key of 5	**15 fewer**	2 points if matched on the first try.
match_s2_4	Set 3: Match half a picture, key of 2	**1**	2 points if matched on the first try.
match_s2_5	Set 3: Match half a picture, key of 10	**5**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: pike place market is in seattle washington	**Pike Place Market is in Seattle, Washington.**	Pike Place Market, Seattle, and Washington are names of places, so each word gets a capital letter. Put a comma between the city and the state, and end with a period.
fs2_s0	Set 1: Fix the sentence: we sold tulips daisies and sunflowers	**We sold tulips, daisies, and sunflowers.** (also accepted: We sold tulips, daisies and sunflowers.)	Start with a capital W, put commas between the flowers in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how many more fish did they sell on saturday	**How many more fish did they sell on Saturday?**	Start with a capital H. Saturday is the name of a day, so it gets a capital S. The sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: arlo beatriz and bodhi made a bar graph	**Arlo, Beatriz, and Bodhi made a bar graph.** (also accepted: Arlo, Beatriz and Bodhi made a bar graph.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: beatriz visited seattle in august	**Beatriz visited Seattle in August.**	Beatriz is a name, Seattle is a city, and August is a month, so each gets a capital letter. End the telling sentence with a period.
fs2_s2	Set 3: Fix the sentence: the three flower stalls was busy	**The three flower stalls were busy.**	Start with a capital T and end with a period. Three stalls means more than one, so use were, not was.

BOSS BATTLE: THE MARKET REPORT (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Use the Saturday shoppers bar graph. How many more shoppers went to the fish market than to the bakery?	**40 shoppers** (also accepted: 40 more, 40 more shoppers, 40)	Fish: halfway between 90 and 100 is 95. Bakery: halfway between 50 and 60 is 55. 95 − 55 = 40 more shoppers.
bb2_s0	Set 1: Find the mistake. Bodhi says the flower stalls and the fruit stand together had 65 more shoppers than the fish market, because 150 − 95 = 65. What went wrong? [He added correctly, 70 + 80 = 150, but subtracted wrong. 150 − 95 = 55, so 55 more shoppers. / Nothing is wrong. It is 65 more shoppers. / The flower stalls and the fruit stand had fewer shoppers than the fish market. / You are not allowed to add two bars on a bar graph.]	**He added correctly, 70 + 80 = 150, but subtracted wrong. 150 − 95 = 55, so 55 more shoppers.**	Flowers 70 + fruit 80 = 150. 150 − 95 = 55. Check: 95 + 55 = 150.
bb1_s1	Set 2: Use the kite picture graph. How many kites were sold in all?	**48 kites** (also accepted: 48)	Dragon 8 × 2 = 16. Fish 5 and a half = 11. Box 4 × 2 = 8. Bird 6 and a half = 13. 16 + 11 + 8 + 13 = 48 kites.
bb2_s1	Set 2: Use the same graph. How many more dragon kites than box kites were sold?	**8 kites** (also accepted: 8 more, 8 more kites, 8, eight, eight kites, eight more, eight more kites)	Dragon: 16. Box: 8. 16 − 8 = 8 more. Or: 4 more pictures × 2 = 8.
bb1_s2	Set 3: Use the cherry bar graph. How many fewer pounds came from Prosser than from Wenatchee?	**25 pounds** (also accepted: 25 fewer, 25 fewer pounds, 25)	Wenatchee: 45 pounds. Prosser: 20 pounds. 45 − 20 = 25 fewer pounds.
bb2_s2	Set 3: Use the same graph. Which two towns together sent exactly 65 pounds? [Wenatchee and Prosser / Yakima and Chelan / Wenatchee and Chelan / Yakima and Prosser]	**Wenatchee and Prosser**	Wenatchee 45 + Prosser 20 = 65. Yakima and Chelan make 60, Wenatchee and Chelan make 70, and Yakima and Prosser make 55.

BOSS BATTLE: CLOSING TIME (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Ms. Akana's goal is 100 bouquets. Her picture graph uses a key where each flower stands for 10 bouquets. It shows 4 flowers for Saturday morning and 3 and a half flowers for Saturday afternoon. How many more bouquets does she need to reach her goal?	**25 bouquets** (also accepted: 25 more, 25 more bouquets, 25)	Step 1: morning 40 + afternoon 35 = 75 bouquets. Step 2: 100 − 75 = 25 more bouquets.
bb4_s0	Set 1: Beatriz made a picture graph where each apple stands for 5 boxes. The Honeycrisp row has 8 apples. The Gala row has 3 fewer apples than the Honeycrisp row. How many boxes of Gala apples does it show?	**25 boxes** (also accepted: 25)	Step 1: Gala has 8 − 3 = 5 apples. Step 2: 5 × 5 = 25 boxes.
bb3_s1	Set 2: Which explanation is correct? Why does the fish kite row show 11 kites? [5 whole pictures show 10 kites, and half a picture shows 1 kite, so 10 + 1 = 11. / There are 5 and a half pictures, so it shows 5 and a half kites. / Half a picture shows 2 kites, so it is 12 kites. / 5 × 2 = 10 kites, and a half picture does not count.]	**5 whole pictures show 10 kites, and half a picture shows 1 kite, so 10 + 1 = 11.**	The key is 2, so half a picture is half of 2, which is 1. 5 pictures × 2 = 10, and 10 + 1 = 11 kites.

ID	Question	Answer	Why (explain to student)
bb4_s1	Set 2: Next week the toy shop wants to sell 20 bird kites. How many more bird kites must it sell next week than it sold this week?	**7 kites** (also accepted: 7 more, 7 more kites, 7 bird kites, 7, seven, seven kites, seven more, seven more kites, seven bird kites)	This week: 6 and a half pictures × 2 = 13 bird kites. 20 − 13 = 7 more kites.
bb3_s2	Set 3: Mr. Bashir buys all the Yakima cherries and all the Chelan cherries. He sells 40 pounds. How many pounds does he have left?	**20 pounds** (also accepted: 20)	Step 1: Yakima 35 + Chelan 25 = 60 pounds. Step 2: 60 − 40 = 20 pounds left.
bb4_s2	Set 3: Find the mistake. Anouk says Chelan sent 50 pounds, because its bar reaches the 5th grid line. What went wrong? [Each grid line counts by 5, not 10. The 5th line is 25, so Chelan sent 25 pounds. / Nothing is wrong. The 5th line is 50. / The bar shows 5 pounds, because it is the 5th line. / The bar shows 30 pounds, because you round to the nearest ten.]	**Each grid line counts by 5, not 10. The 5th line is 25, so Chelan sent 25 pounds.**	Read the scale: 5, 10, 15, 20, 25. The 5th grid line is 25 pounds.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W21.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: bakery opens 6:50 a.m.; loaves out 35 minutes later	7:25 a.m. Jumps: 6:50 to 7:00 is 10 minutes, then 25 more minutes to 7:25.
2	Key: each flower = 2 bouquets; 5 pictures	10 bouquets (2, 4, 6, 8, 10).
3	Grid lines count by 10; bar halfway between 20 and 30	25
4	Part that tells how many each picture stands for	the key
5	Picture graph row for 15 apples, each circle = 5 apples	3 circles (15 ÷ 5 = 3). Accept any reasonable attempt before the lesson.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Key of 10: plums 4 pictures, pears 2 and a half pictures	15 more. Plums 40 boxes, pears 25 boxes, and 40 − 25 = 15.
2	Bar graph, scale of 5: Clover 20, Fireweed 35, Blackberry 10	Grid lines labeled 0, 5, 10, 15 ... at least to 35. Clover bar stops on 20, Fireweed bar on 35, Blackberry bar on 10. Check for a title and labels.
3	Lines count by 2; Bodhi says the 4th line shows 4 kites	He counted the lines instead of using the scale. The lines count 2, 4, 6, 8, so the 4th line shows 8 kites.
4	Each star = 5 visitors: Monday 6 stars, Tuesday 9 stars; goal 100	Step 1: 6 × 5 = 30 and 9 × 5 = 45, so 30 + 45 = 75 visitors. Step 2: 100 − 75 = 25 more visitors.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Why a key of 10 works for 30, 50 and 80 tulips but a key of 20 does not	30, 50 and 80 are all counts by 10, so the key of 10 gives 3, 5 and 8 whole pictures. 30 and 50 are not counts by 20, so a key of 20 would need half pictures (1 and a half and 2 and a half).